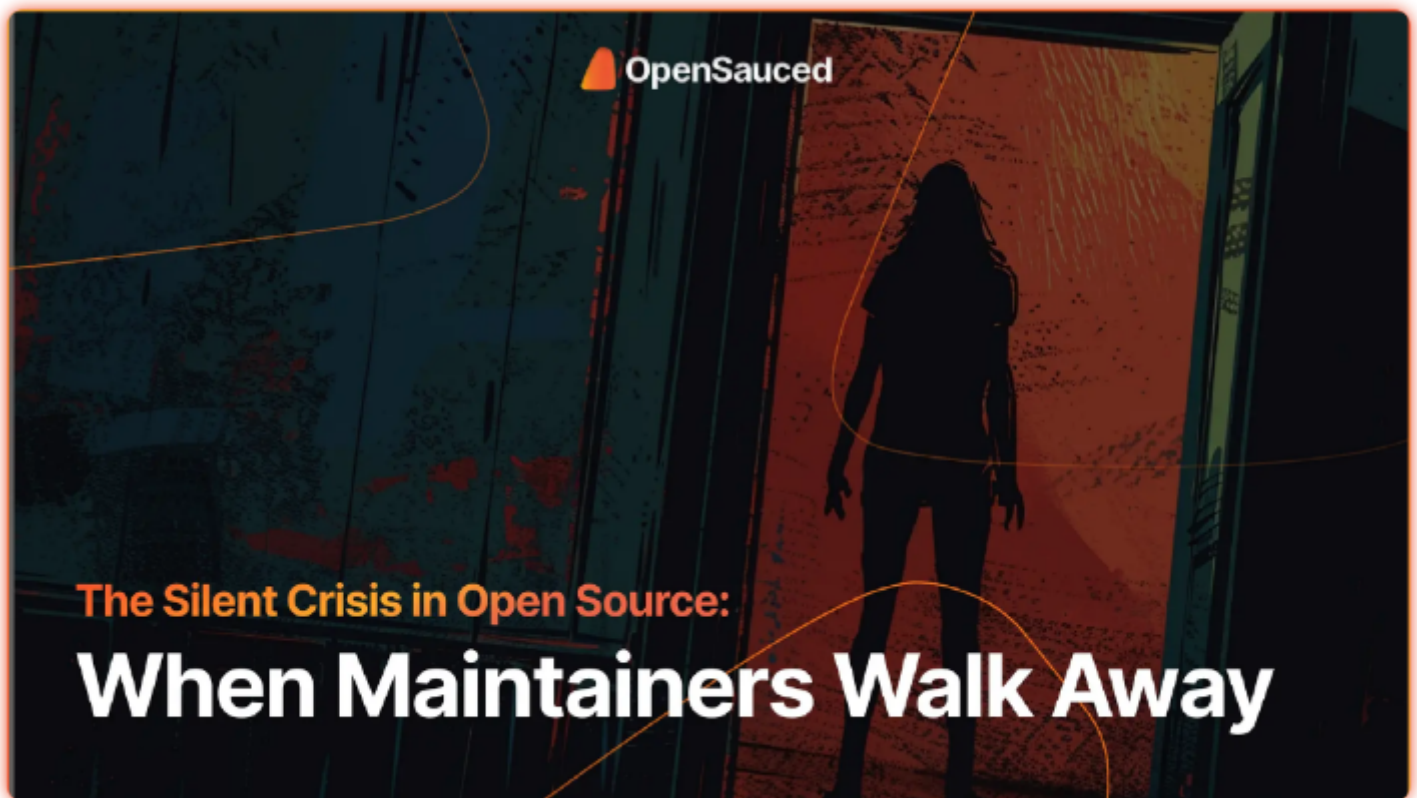


#maintainers

# The Silent Crisis in Open Source: When Maintainers Walk Away

 BekahHW  7 mins read  2024-07-03



In May 2022, Dane Springmeyer, the primary maintainer of `node-pre-gyp`, a critical tool in the Node.js ecosystem, announced his decision to step down. This wasn't just another developer moving on; for nearly a decade he had been maintaining the project.

Despite outlining the urgency and the need for community involvement to keep the project maintained, proposing several options for the future of the project, and emphasizing the importance of maintaining or gracefully deprecating `node-pre-gyp`` to avoid disruptions, it wasn't resolved until very recently.

This situation is just one that captures the challenges of maintainer transitions in open source projects. The departure of a key maintainer can have far-reaching implications, affecting the project's sustainability, security, and overall health.

## The Impact of a Maintainer's Departure

Unaddressed pull requests and unresolved issues start piling up, creating a backlog and uncertainty for the contributor community. The lack of regular updates can expose the project to security vulnerabilities, quickly decreasing trust among users and contributors. For ``node-pre-gyp``, this scenario unfolded as community members scrambled to find a way forward. **Single points of failure can have widespread repercussions.**

There's another type of loss that we forget about: the loss of institutional knowledge and context. Maintainers often possess critical insights into the project's history, design decisions, and future roadmap. Without this knowledge, new maintainers may struggle to navigate the project effectively. This loss of continuity can disrupt the project's development and direction, impacting its long-term viability.

This is one of the reasons, we make the **lottery factor** visible on repository pages on OpenSauced.

*"The Lottery Factor is a metric that identifies how at risk a project is if a key contributor leaves. It is calculated by the percentage of pull request (PR) contributions made by the top contributors. If 50% of the PR contributions come from two or fewer contributors, the lottery factor is high."*

Understanding how at risk a project is if a key contributor leaves can help contributors and maintainers prepare for potential transitions and ensure the project's

sustainability.

# The Invisible Maintainer: A Hidden Threat

One often overlooked aspect of the maintainer crisis is the difficulty in identifying who actually maintains a project. This lack of transparency can lead to communication breakdowns, unclear decision-making processes, hidden lottery factor, and accountability issues.

Understanding who the maintainers are is key to project health. If you take a look at the `node-pre-gyp` contributors dashboard, you'll see that they were able to make the transition to a new maintainer.



This isn't just about giving credit where it's due. It's about supporting the community with necessary intelligence.

When we know who the maintainers are, we:

- Know who's behind our dependencies.
- Can target our open source support. Find the overworked maintainers and back them up.
- Can actively support the next generation of project leaders.

It's one step in the right direction, but it's not enough. We need to make maintainer transitions more transparent, predictable, and manageable.

## The Ticking Time Bomb of Maintainer Burnout

Springmeyer's case is far from unique. Across the open source ecosystem, projects are vulnerable to maintainer abandonment for a variety of reasons, including burnout, loneliness, or lack of support. In the case of springmeyer, he cited personal and professional shifts, such as parental leave and changing priorities at Mapbox. Marak, the Faker.js creator, intentionally deleted Faker.js to highlight the pressures and lack of open source support maintainers often face. These situations emphasize the need for a supportive infrastructure that recognizes and alleviates the burden on maintainers.

This current unsustainable model is a ticking time bomb. When key maintainers leave, projects can quickly become outdated, insecure, or completely non-functional. The ripple effects can be catastrophic, potentially impacting thousands of dependent projects and millions of users.

## The Trust Paradox: The Hidden Challenge of Maintainer Succession

Finding new maintainers isn't just about identifying skilled developers. It's about trust. Handing over the keys to a project with millions of downloads to someone else can be risky. This trust paradox creates a dilemma:

*""[If] you are maintaining a project, you're kind of a gatekeeper. If you oftentimes you don't want to be, but also you don't have the time to onboard some random people, because you're afraid that like the millions of downloads this package has will fall into the hands of someone you don't know, and they could really cause damage, right? So how to do that?" - Gregor Martynus on The Secret Sauce"*

This quote identifies a critical dilemma:

- Maintainers may find themselves reluctant guardians of widely-used projects.
- The time investment required to properly vet and onboard new maintainers is substantial.
- Giving someone maintainer access to a project can carry enormous risk.

This creates a vicious cycle: overworked maintainers struggle to find time to onboard help, leading to further burnout and increasing the risk of sudden project abandonment.

## Quantifying the Risk: New Metrics for Project Health

To address this problem, we need better ways to assess the health and sustainability of open source projects. Two emerging metrics offer valuable insights:

1. **Lottery Factor:** This metric highlights the dependency on key contributors. A project is considered vulnerable if 2 or fewer contributors account for 50% or more of the project's contributions. It's essentially measuring the "bus factor" - how many people could the project lose before it's in serious trouble?
2. **Contributor Confidence:** This metric predicts the likelihood that users who star or fork a repository will return to make contributions. A higher score (typically 30-40%) indicates a healthy, active project where contributions are valued.

These metrics offer a data-driven approach to assessing project health. A high Lottery Factor combined with low Contributor Confidence could be a red flag, indicating a project overly reliant on a small number of contributors and struggling to attract new ones.

## Preventing the Exodus: A Call to Action

1. **Sustainable Funding Models:** We need to move beyond the "free labor" mindset. Companies benefiting from open source projects should contribute financially, either through direct sponsorship or by allocating paid developer time to maintenance to prove open source support.
2. **Succession Planning:** Projects should actively cultivate a bench of potential maintainers. This means documenting processes, sharing knowledge, and actively supporting new contributors. Aim to lower that Lottery Factor!
3. **Community Health Metrics:** Regularly track metrics like Contributor Confidence. If it starts to dip, it may be time to invest in community outreach or simplify the contribution process. Healthy communities attract and retain contributors.

These approaches don't just make maintainer succession safer—they make it more achievable. By reducing the cognitive load of vetting new maintainers, we lower the barrier to expanding the maintainer pool.

# Redefining Open Source Insights

The departure of Dane Springmeyer from `node-pre-gyp` wasn't just a personal decision — it was a reminder of the hidden fragility of our digital infrastructure.

We need to move into a new era in open source; one where gut feelings and GitHub stars aren't seen as metrics of project health. One where the true pulse of a project — its maintainer dedication, community activity, and sustainability — can be quantified and understood.

We should strive to:

- See beyond commit counts to understand the real dynamics of contributor engagement and open source support
- Identify the unsung heroes holding projects together
- Predict potential maintainer burnout before it happens
- Understand the true lottery factor of your essential dependencies

This should be the reality of open source intelligence. When we adopt this mindset, we can:

- Make informed decisions about which projects to rely on and support
- Focus contributor efforts where they'll have the most impact
- Proactively address project vulnerabilities
- Rally around projects in need before they reach crisis point

This vision requires a shift in mindset. We have to move beyond simple metrics and anecdotal evidence and embrace data-driven insights that reveal the true health of



our open source ecosystem. It's time to truly see the human element that drives open source forward.

It's about making the invisible visible. It's about transforming raw data into actionable intelligence that ensures open source software support and sustainability.



## BekahHW

Bekah graduated from a coding bootcamp in May of 2019 and since then has spent time as a frontend developer, started the Virtual Coffee tech community, spent time in DevRel and has continued to mom her four kids. She currently co-hosts the Compressed.fm and Virtual Coffee podcasts, lifts heavy things in her free time, & works as the Developer Experience Lead at OpenSauced.

### RECENT POSTS



## OpenSauced is joining the Linux Foundation

 bdougie  2 mins read



OpenSauced joins Linux Foundation, making AI-powered open source analytics freely available while expanding beyond GitHub to serve the broader open so...

[Read More](#)



#kubernetes

## OpenSauced on Azure: Lessons learned from a near-zero downtime migration

 John McBride  5 mins read  2024-10-15

How the OpenSauced engineering team made a near-zero downtime migration to Microsoft Azure

[Read More](#)